Q1. Define a relation S on Z as follows: For all integers a and b (8 marks)

a T b $\leftrightarrow$ 3 | (a^3 - b^3)

a. Is 1 S 4?

1^3 - 4^3 = -63 and 3 | -63

Answer : Yes

b. Is 2 S (-1)?

2^3 - $(-1)^3$ = 9 and 3 | 9

Answer : Yes

c. Is 3 S 10?

3^3 - 10^3 = -973 and 3 | -973

Answer : No

d. Is (-5) S 5?

$(-5)^3$ - 5^3 = -250 and 3 | -250

Answer : No

Q2. Let A = {0, 1, 2, 3}. For each relation below:

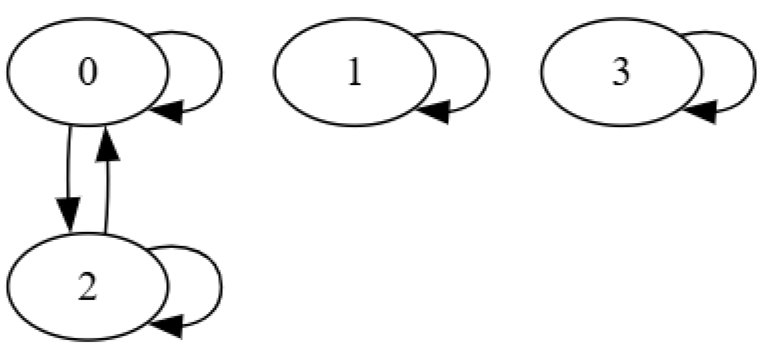
a. Draw the directed graph. (5 x 3 = 15)

b. Determine if it is Reflexive, Symmetric, or Transitive. (3 x 3 = 9)

c. Provide counterexamples for any properties that fail. (1 x 3 = 3)

1. R1 = {(0, 0), (1, 1), (2, 2), (3, 3), (0, 2), (2, 0)}

(a) Directed Graph :



(b) Properties :

Reflexive : Yes, because every element has a loop (a,a) $\in$ R1.

Symmetric : Yes. Check: $(0,2) \in R1$ and $(2,0) \in R1$; all loops are symmetric

Transitive : Yes. The only non-loop pairs are between 0 and 2.

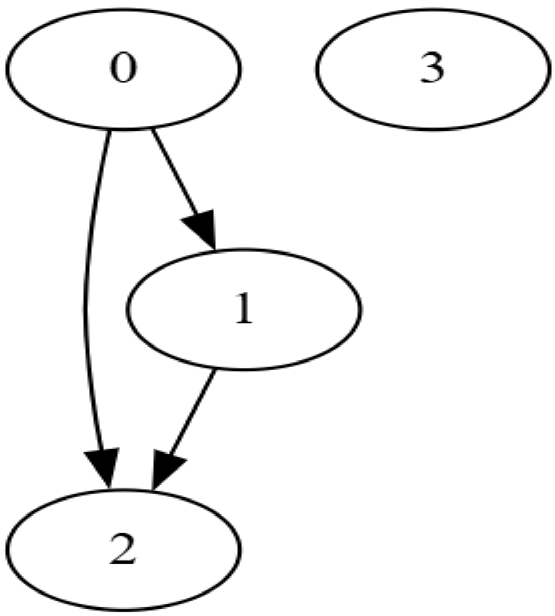
(0,2) and (2,0) —> (0,0) in R1.
(2,0) and (0,2) —> (2,2) in R1.   All other conditions are satisfied trivially.
So transitivity holds.

(c) Counterexamples :

None – all properties hold.

2. R2 = {(0, 1), (1, 2), (0, 2)}

(a) Directed Graph :



(b) Properties :

Reflexive : No, Counterexample: (0,0) not in R2, also (1,1),(2,2),(3,3) missing.

Symmetric : No, Counterexample:

(0,1) in R2 and (1,0) is not in R2.
(1,2) in R2 and (2,1) is not in R2.
(0,2) in R2 and (2,0) is not in R2.

Transitive : Yes.

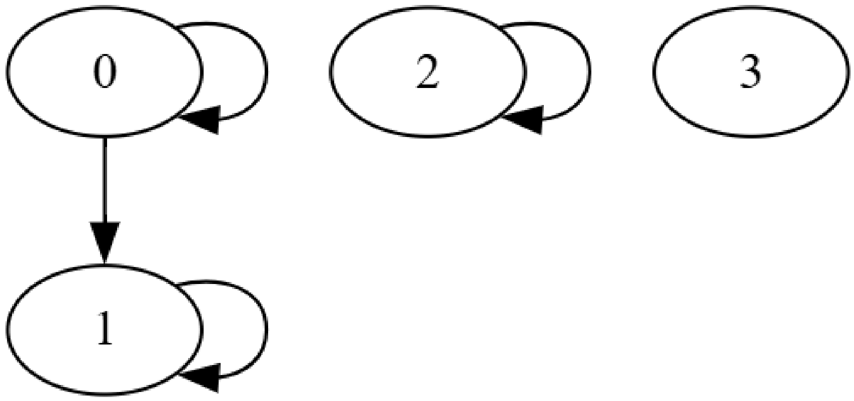
(0,1) and (1,2) —> (0,2) in R2.
So transitivity holds.

(c) Counterexamples :

Only reflexivity and symmetry fails; for transitivity no counterexamples exist.

3. R3 = {(0, 0), (0, 1), (1, 1), (2, 2)}

(a) Directed Graph :



(b) Properties :

Reflexive : No, Counterexample: (3,3) not in R3.

Symmetric : No, Counterexample:

(0,1) in R2 and (1,0) is not in R3.

Transitive : Yes.

(0,1) and (1,1) —> (0,1) in R3.

(0,0) and (0,1) —> (0,1) in R3.

So transitivity holds.

(c) Counterexamples :

Only reflexivity and symmetry fails; for transitivity no counterexamples exist.